

# Econ 101A

## Section 9: Midterm Review

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This is a review, not new material. The goal is to see the whole unit as one connected picture and to close out a few problems we did not finish in section. Nothing here is meant to be hard. If you can follow the map below and redo the short problems, you are in good shape. Take a breath, you know more than you think.

### 1 The Map: One Consumer Problem, Many Faces

Everything this unit fits on one page.

**1. Preferences.** Rational means *complete* (you can compare any two bundles) and *transitive* (no cycles). A utility function is just a numbering of bundles that respects the ranking. Any strictly increasing transform of  $u$  (like  $e^u$  or  $\ln u$ ) represents the same preferences and gives the same demands.

**2. Utility maximization.** Choose the best affordable bundle:

$$\max_{x,y} u(x,y) \text{ s.t. } px + qy = M, \quad \mathcal{L} = u + \lambda(M - px - qy).$$

Interior optimum: tangency  $u_x/u_y = p/q$ , plus the budget. Watch for the traps: perfect substitutes and other corners, and non-convex preferences where the tangency is a minimum (check a corner or the second-order condition).

**3. Marshallian demand and indirect utility.** Solving gives  $x^*(p, q, M)$  and  $v(p, q, M) = u(x^*, y^*)$ . By the envelope theorem,  $\partial v / \partial M = \lambda$  (marginal utility of income) and  $\partial v / \partial p = -\lambda x^*$ .

**4. The dual: expenditure minimization.** Same tangency, different constraint:

$$\min px + qy \text{ s.t. } u(x, y) \geq \bar{u} \Rightarrow \text{ Hicksian } h(p, q, \bar{u}), \quad e(p, q, \bar{u}) = ph_x + qh_y.$$

Shephard:  $\partial e / \partial p = h_x$ . Duality:  $h_x(p, q, \bar{u}) = x^*(p, q, e)$ .

**5. Slutsky.** The total price effect splits into a substitution part (always  $\leq 0$ ) and an income part:

$$\frac{\partial x^*}{\partial p} = \frac{\partial h_x}{\partial p} \bigg|_{\bar{u}=v} - x^* \frac{\partial x^*}{\partial M}.$$

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Normal good: both terms negative, demand slopes down. Giffen: needs an inferior good with an income effect large enough to flip the total positive. Every Giffen good is inferior; not every inferior good is Giffen.

**6. Applications.** Same tools, relabeled. Labor supply: leisure priced at the wage  $w$ , full income  $M + wT$ , tangency  $u_\ell/u_c = w$ , reservation wage below which hours are zero. Intertemporal: future consumption priced at  $1/(1+r)$ , wealth  $W = M_0 + M_1/(1+r)$ , Euler  $U'(c_0) = \frac{1+r}{1+\delta}U'(c_1)$ , smoothing when  $r = \delta$ .

## 2 Warm-up Problems

### Problem 1 (preferences and transforms)

A consumer's preferences are represented by  $u(x, y)$ . Does  $\tilde{u} = e^u$  represent the same preferences? Would maximizing  $\tilde{u}$  subject to the same budget change the optimal bundle?

### Problem 2 (perfect substitutes)

Let  $u(x, y) = 2x + y$  with prices  $p, q > 0$  and income  $M$ . Find the optimal bundle and say when it is not unique.

### Problem 3 (Cobb–Douglas quick pass)

Let  $u(x, y) = \sqrt{xy}$ , prices  $p, q$ , income  $M$ .

- (a) State the Marshallian demands.
- (b) Is  $x$  normal? Is it a gross substitute or complement for  $y$ ?

### Problem 4 (Giffen versus inferior)

Using the Slutsky equation, explain why a Giffen good must be inferior, and why an inferior good need not be Giffen.

### Problem 5 (numerical Slutsky)

Keep  $u = \sqrt{xy}$  at  $p = q = 1$ ,  $M = 8$ , where  $x^* = M/2p$ ,  $y^* = M/2q$ , and the Hicksian is  $h_x = \bar{u}\sqrt{q/p}$ . Decompose  $\partial x^*/\partial p$  into substitution and income parts.

### Problem 6 (applications one-liners)

- (a) In the labor model with non-labor income  $M$  and endowment  $T$ , what is the reservation wage, and what happens below it?
- (b) In the two-period model with  $r = \delta$ , what is the relation between  $c_0$  and  $c_1$ , and how do you know whether the consumer saves?

**One last thing.** If a problem looks unfamiliar, do not panic. Write the budget, write the Lagrangian, take first-order conditions, and read off what the question asks. The steps are always the same. Good luck, you have got this.